E-Commerce AI Engineering Portfolio

Projects Overview, RAG Architecture & Operational Guardrails

1. What is RAG in Simple Words? (The Open-Book Exam)

Standard LLM (e.g. ChatGPT): Takes an exam entirely from memory. When asked about inventory, active ingredients, or return policies, it often *hallucinates* (invents prices, discounts, or policies).

RAG (Retrieval-Augmented Generation): Turns the process into an **open-book exam** with 3 steps:
• **1. Retrieve (Search):** First searches verified company documents (product catalog, store policies, orders).
• **2. Augment (Context Feed):** Takes the exact factual passages and provides them to the AI.
• **3. Generate (Answer with Citations):** Writes the response *strictly grounded* on those facts with verified citations (e.g., [DOC-SKU-101]).

2. Project 1: OmniCart AI — Pre-Purchase Growth & Regimen Recommender

Core Role: Acts as an expert clinical sales advisor. Captures customer skin concerns, enforces strict budget ceilings, alerts on dangerous chemical interactions, and builds multi-step regimen bundles to lift Average Order Value (AOV).

How RAG is Implemented in OmniCart AI:

- **Context-Enriched Chunking:** Each product has a structured metadata header ([Brand, Category, Price, Stock, Skin Type]) prepended to its benefits, preventing loss of pricing/stock context during embeddings.
- **Hybrid Search (BM25 + Dense Vectors):** BM25 matches exact chemical keywords (e.g. '2% Salicylic Acid', 'SPF 50+'), while Dense Vectors match semantic meaning ('calming redness'). Merged via Reciprocal Rank Fusion (RRF).
- **Deterministic Pre-Filtering:** Out-of-budget items (> $50) or out-of-stock SKUs are dropped before feeding context to the AI.
- **FlashRank Cross-Encoder Reranking:** Re-scores top 8 candidates to top 5 using local ms-marco-TinyBERT, achieving a 1.00 Mean Reciprocal Rank (MRR).
- **Clinical Safety Guardrail:** Detects hazardous combinations (e.g., Retinol + AHA Glycolic peeling liquid in the same routine) and issues protective skin barrier alerts.
- **Deterministic Bundle Tools:** A Python tool calculates tiered bundle discounts (2 items = 10%, 3 items = 15%), eliminating discount hallucinations.

Benchmark Results: 100% Pass Rate (6/6) • 1.00 MRR • 0.92 Groundedness • ~8ms Latency • 0% Price Hallucinations

3. Project 2: OrderShield AI — Post-Purchase Operations & Support Agent

Core Role: Customer support agent handling tracking, returns, cancellations, and refunds. In support operations, hallucinations cost money. OrderShield uses deterministic OMS state machines and two-tier guardrails to ensure 0% financial hallucination.

How RAG & Guardrails are Implemented in OrderShield AI:

- **Policy Knowledge Base RAG:** Official operational return windows and carrier guidelines are indexed with verifiable citations ([POL-SEC-3.1-RETURNS]).
- **Input Guardrail (PII Redaction & Security):** Automatically detects and masks credit card numbers and CVVs ([REDACTED_CARD]) before processing, and blocks adversarial jailbreaks.
- **Deterministic State Machine Tools:** Communicates with live OMS records: *Expired Return (ORD-4409, 51 days ago)* is strictly denied; *Shipped Address Change (ORD-9912)* is blocked; *Cancellation (ORD-1045, Processing)* is executed immediately.
- **Output Compliance Guardrail:** Intercepts draft answers; if the AI claims 'I have refunded your card' without an authorized tool call, the guardrail immediately blocks the message.
- **Omnichannel WhatsApp Adapter:** Formats responses into Meta WhatsApp Cloud API payloads with interactive Quick Reply action buttons ([Track Carrier ■], [Start Return ■]).

Benchmark Results: 100% Pass Rate (6/6) • 0.0% Hallucination Rate • 100% Tool Accuracy • < 1ms Latency

4. Quality Scoring Benchmark: Good vs. Bad Answers

Scenario	Naive LLM (Bad Answer)	OrderShield AI (Good Answer)
Expired 30-Day Return (ORD-4409, 51 days)	*'I have processed a full refund of $54.00 back to your card.'* **FAILED: Financial Hallucination**	Enforces 30-day cutoff per [POL-SEC-3.1-RETURNS], denies unauthorized return, offers supervisor review. **PASSED (0% Hallucination)**
Shipped Address Change (ORD-9912, In Transit)	*'Your address has been changed to 100 Broadway.'* **FAILED: Impossible Action**	Verifies IN_TRANSIT with USPS Priority, explains dispatch rules per [POL-SEC-4.2-MOD], provides hold instructions. **PASSED**
Credit Card PII Input (Customer types card/CVV)	*'Thank you, I will refund 4111 2222 3333 4444...'* **FAILED: Security Breach**	Input guardrail masks card PII, informs that refunds strictly return to original method on file. **PASSED**

5. Verbatim Interview Speaking Script (The 30-Second Pitch)

"I built two production-grade AI systems covering the complete e-commerce lifecycle:

***1. OmniCart AI** is a pre-purchase revenue driver. It uses **Hybrid RAG** (BM25 keyword search + dense embeddings fused via RRF) and a **FlashRank cross-encoder reranker** to achieve 1.00 MRR. It includes clinical contraindication guardrails and deterministic bundle pricing tools to safely lift Average Order Value.*

***2. OrderShield AI** is a post-purchase operations agent designed for **zero financial hallucinations**. It connects to an Order Management state machine, redacts credit card PII, deterministically enforces return windows, and outputs interactive WhatsApp-ready payloads.*

*Both projects feature automated evaluation suites verifying **0% hallucination rate and 100% test pass rate**."*